Research Program

Research Program Title:

Neural circuits for somatosensation from mice to naked mole-rats

Research Abstract:

The soft caress from a loved one evokes a rewarding emotional response, whereas a mosquito bite triggers unpleasantness and aversion. As a field, we have learned much about how we detect a myriad of somatosensory stimuli on the skin. However, a major gap resides in understanding the integrated response from the periphery to the brain, which hinders our understanding of why we crave pleasurable touch while avoiding noxious insults.

In one major research direction, we will build on our expertise in measuring both the sensory and emotional dimensions of pain, and use longitudinal multimodal imaging approaches, measuring both behavior and brain activity – all delineated by the type of chronic pain. This work will allow us to predict the trajectory of pain chronicity, across sex and lifespan, producing an expansive atlas of behavior and neural signatures associated with distinct types of chronic pain – facilitating much needed breakthroughs in the field.

In a second major research direction, we will move below the surface (quite literally); studying an animal that has evolved for life underground in an exquisite eusocial hierarchy. The naked mole-rat from East Africa has extremes in somatosensation – displaying insensitivity to some forms of pain while paradoxically having a vastly expanded somatosensory cortex, suggesting that these animals should be highly sensitive tactile detectors. Based on new data from my lab with naked mole-rats, I hypothesize that the extremes in somatosensory behavior and cortical representation are linked to eusociality, perhaps driving the animal's highly social structure. Learning how ~35 million years of evolution has shaped the nervous system of the naked mole-rat for an extreme lifestyle could provide fundamental knowledge about how the nervous system of other mammals could be altered for benefit, such as relief from pain or even driving pro-social behavior.

Neural circuits for somatosensation from mice to naked mole-rats

Background and Significance

Behavioral and neural circuit mapping of highly complex pain states

Chronic pain is a complex state driven by injury, infection, or disease that negatively alters our sensory experience of the world and our internal emotional states. The measurement tools that have historically been used in the field provide limited degrees of resolution, and chronic pain is often lumped together as a single phenomenon. The lack of delineation and resolution of chronic pain by its origin, ranging from inflammatory pain to neuropathic pain to bone cancer pain, for example, has limited our fundamental understanding of peripheral and central mechanisms that drive specific types of pain. My lab is tackling this problem using a combination of high-speed videography, statistical modeling, automated limb tracking with deep neural networks, and custom software [1-4].

With an unprecedented resolution to quantify many dimensions of pain in mice, what big problems in pain science are we now poised to address? One of the biggest unresolved mysteries in the field is how concerted patterns of neural activity across the brain generate pain sensation versus pain emotion, and how these networks emerge as chronic pain develops, and how these networks are unique depending upon pain type or underlying genetic differences. Using the traditional lab mouse and a powerful model that I have started to establish in my laboratory, the naked mole-rat, I will seek to go after this problem as a Freeman Hrabowski Scholar and will outline novel approaches below in the *Future Directions* section to go after this challenge.

Pain sensation and social structure in the naked mole-rat

The naked mole-rat hails from East Africa and has adapted to life underground in a highly cooperative "eusocial" structure with one breeding queen, one or a few breeding males, and subservient caste members [5]. A single colony can contain a few dozen or a few hundred animals with members living upwards of 30 years [6]. Most directly related to my lab's interests in pain, the naked mole-rat does not sense inflammation-induced pain. Peripheral alterations, ranging from reduced numbers of C-fiber nociceptors, gene polymorphisms in pain sensing ion channels, and a lack of pain-linked neuropeptides contribute to the animal's lack of response to normally painful molecules like capsaicin, acid, and low pH [7,8]. Paradoxically however, naked mole-rat nociceptors do fire normally to capsaicin in vitro in a TRPV1 dependent manner[7], although no behavioral response is triggered in vivo, suggesting other mechanisms at play for pain inhibition, that perhaps reside centrally. The role of the brain in mediating pain inhibition in the naked mole-rat is completely unexplored, while pain inhibition from the brain is a powerful system, described and studied since the 1970s [9-13]. At a mechanistic level, most of the unusual and extreme traits of the naked mole-rat remain unknown or poorly defined because these animals have been understudied, especially compared to standard lab mice and rats. Determining how the brain might control any of these traits, remains almost completely unknown, as no lab has ever recorded from their brains in awake animals.



Figure 1 XXXXXXXXXX naked mole-rat colony housed at XXXXXX XXXXXXXXXXXXXXXXX. Note the interconnected colony system. We maintain the room temperature at 80°F and 50% humidity to mimic conditions in East Africa.

Animals that lie at the extremes of biological specialization, like the naked mole-rat, offer an opportunity to uncover novel solutions to solving problems that evolution has randomly stumbled upon. Shortly after arriving at XXXXX University nearly one year ago, I started a naked mole-rat colony and we are underway collecting pilot data (**Figure 1**). As a Freeman Hrabowski Scholar I plan to open up the naked mole-rat as a powerful model to perform modern neuroscience

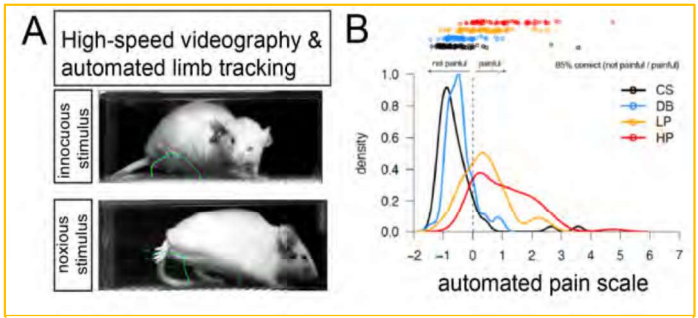


Figure 2. A) Using high-speed videography (2,000 frames/second) with deep neural network based paw tracking, responses to innocuous stimuli are simple "up-down" withdrawals, whereas noxious stimuli evoke more elaborate withdrawal patterns. This distinction is missed with the naked eye. **B)** The univariate linear scale that best separates the four pain states studied: CS, DB, LP, and HP (CS,DB=innocuous stimuli, LP,HP=noxious stimuli). Kernel density estimates and the original data decomposed by pain state are shown projected onto the univariate measure of pain identified by ordinal logistic regression. Here we use the threshold between non-pain and pain (DB and LP) inferred by ordinal logistic regression as the zero point, and we rescale by the threshold between low pain (LP) and high pain (HP), fixing this point at one. Adapted from XXXXXXXXXXXXXX.

research and comparative genomics. This is an ambitious undertaking that is too premature for federal funding or other private foundations, but the high-risk high-reward proposition that these studies offer, is ideal for HHMI support at this stage of my career.

Prior and Active Projects

Creation of "pain scales" to quantify pain

An innovative aspect of our work is recording sub-second behaviors in freely behaving rodents and analyzing the data with artificial intelligence (AI) to measure pain [1-4]. In recent senior authored papers from my lab, we developed rodent "pain scales" (**Figure 2**). This automated technology, that we named PAWS (Pain Assessment at Withdrawal Speeds), allowed us to measure pain sensation versus pain "affect" in the mouse in an unbiased manner. Additionally, our tools are open source and we have written a protocol paper to allow other users to easily adopt this technology [2].

The role of sensory neurons in rewarding and therapeutic social touch

Although the field has identified sensory neurons that detect discriminative touch, the neurons in the skin that detect social touch and shape the valence of perception generated in the brain, remain unknown. This gap in knowledge is critical, especially when considering the nature of neurodevelopmental disorders like autism spectrum disorder, where gentle touch and socially rewarding behaviors are aversive [14,15].

We began our focus on a population of sensory neurons labeled by the G-protein coupled receptor Mrgprb4 that detects stroking touch in mice [16,17]. We studied the social relevance of Mrgprb4-lineage neurons by genetically engineering mice to allow activation or ablation of this population and revealed that these neurons are required for sexual receptivity and sufficient to induce dopamine release in the mesolimbic reward pathway (**Figure 3**). Even in social isolation, optogenetic stimulation of Mrgprb4-lineage neurons through the back skin is sufficient to induce a conditioned place preference (**Figure 4**) and a striking dorsiflexion resembling the lordotic copulatory posture in females (**Figure 5**). Collectively, our emerging studies support the hypothesis that activating Mrgprb4-lineage neurons in the skin drives a neural system in the brain that encodes the rewarding nature of touch during social behaviors (XXXXXXXXXXXXXXXXXXXXXXXXXXX XXXXXXXXXXX).

In a related paper led by another graduate student in my group, XXXXXXXXXXXXXXXXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXXXXXXXXXXXX, we demonstrated that Mrgprb4-lineage neurons are required to promote stress resilience (**XXXXXXXXXXXXXXXXXXXXXXXXXX** XXXXXXXXXXX). To gain insight into where in the brain alterations maybe occurring, driven by our peripheral manipulations, we used chronic multi-circuit neurophysiological recordings simultaneously in 7 brain regions implicated in reward and stress, in collaboration with Dr. XXXXXXXXXX (**Figure 6**). We used a phase offset approach to estimate signal directionality between each pair of brain regions for both *Mrgprb4^{Cre}; Rosa^{DTA}* mice and controls. At 1-8Hz, the NAc was leading by ~10-15ms +/- 3ms in both genotypes. At 9-10Hz, the VTA is

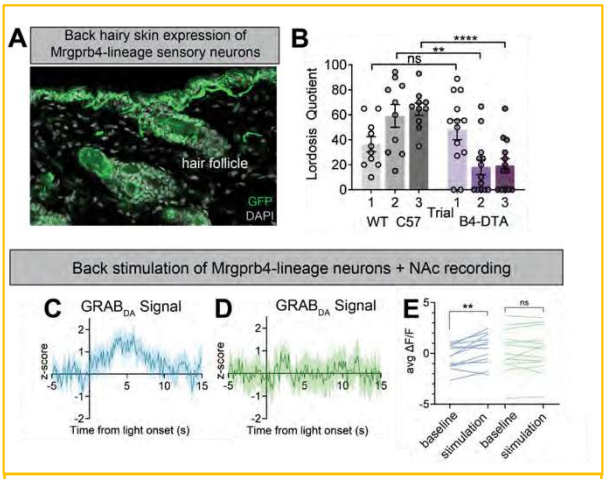


Figure 3. Mrgprb4-lineage neurons are required for sexual receptivity and trigger dopamine release. A) Immunostaining through the skin of transgenic animals showing Mrgprb4-lineage sensory neurons. B) Lordosis quotient scores (proxy for sexual receptivity) for 3 sequential trials for WT C57 and *Mrgprb4^{Cre}; Rosa^{DTA}* females. (**P<0.05, **P<0.01, ***P<0.001,* One-Way ANOVA). B4-DTA represents B4-lineaga ablated mice. Each circle is an individual mouse. C,D) Average GRAB$_{DA}$ delta F/F signals for green (control) and blue light. E) Average deltaF/F pre stim (-5-0) subtracted from average deltaF/F post stim (10-15s) is significantly greater for blue light stimulation compared to the green light control, (**P<0.05,* unpaired t-test). N=10-15 mice for each laser light. Adapted from XXXXXXXXXXXXXXXXXX

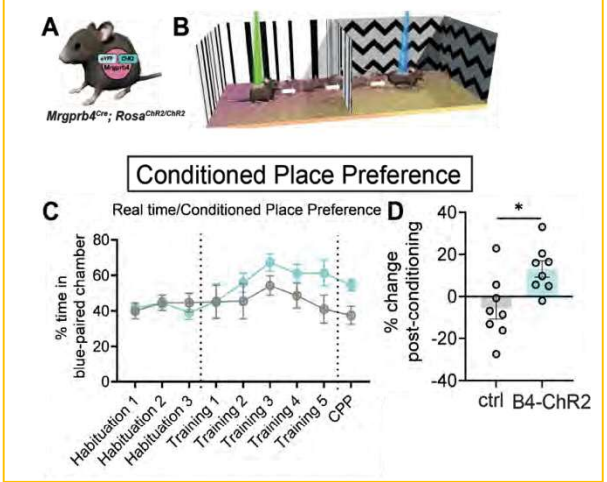


Figure 4. Focalized activation of Mrgprb4-lineage neurons in the back skin is rewarding. A,B) Schematics illustrating transgenic mouse and our real time/conditioned place preference assay. C) *Mrgprb4^{Cre}; Rosa$^{ChR2/ChR2}$* females gradually spend more time in the blue laser chamber compared to green laser chamber during the training days (lasers on) and D) spend significantly more time in the chamber they learned to associate with blue light on the test day (lasers off) compared to habituation days, (**P<0.05,* unpaired t-test.) Adapted from XXXXXXXXX XXXXXXXXXXXXX.

leading by 41ms +/- 5.5ms in control mice but by only 16ms+/- 2.6ms in *Mrgprb4^{Cre}; Rosa^{DTA}* mice (**Figure 7**). These results suggest that an alteration in reward processing is linked to the stress susceptibility phenotypes – connecting neurons in the skin to circuitry and behaviors driven by the brain.

Mapping somatosensory behaviors in the naked mole-rat in groups vs. isolation

Quite peculiar amongst the mammalian kingdom, the naked mole-rat does not appear to respond to capsaicin, acid, or low pH – normally painful stimuli for most mammals [7,8]. Paradoxically however, the animals have a 3-fold expanded somatosensory cortex compared to other mammals [18], suggesting that they should be highly sensitive detectors of somatosensory stimuli and not insensitive to such stimuli. Relatedly, naked mole-rats have whisker-like protrusions that tile their entire bodies, further bolstering the idea that their somatosensory world might be heightened. We decided to investigate this. Stemming from our expertise in mapping somatosensory behaviors in mice and rats with high-speed videography [1,3,4,19], we turned this approach towards the naked mole-rat and made some interesting observations.

To perform our high-speed behavioral mapping to somatosensory stimuli, we typically isolate individual animals and record their responses to given stimuli. Our naked mole-rat colony consists of ~50 animals and they all live in an interconnected cage and tunnel system (**Figure 1**). Strikingly, we noticed that when we touched the isolated naked mole-rats on their face with a soft make-up brush, the animals had a very fast retraction away from the stimulus, coupled to adopting a fighting posture and loud vocalizations (**Figure 8A**). The animals responded to a soft stimulus as painful or aversive. *How solitude profoundly alters the somatosensory of naked mole-rats is unknown and it is one of the questions that we will go after by studying their brains in isolation or in groups*. When we performed the same experiment right after adding one naked mole-rat from the colony into the testing chamber, the stimulated naked mole-rat now had a normal touch response to the stimulus (**Figure 8B**). Of note, the touch of another naked-mole rat to the previously isolated animal did not cause an aversive response, suggesting that they might be interpreting the soft brush as an external threat and not another naked mole-rat. Additionally, as soon as the companion naked mole-rat was removed and we tested the experimental animal again, the painful/aversive response immediately returned (**Figure 8B**). We performed the exact same experiments in the dark with standard lab mice and we did not observe the response we witnessed with naked mole-rats (**Figure 8C**). ***Together, these results demonstrate that it hurts, or at least it is aversive, for naked mole-rats to be touched while alone and away from their colony members.*** It is tantalizing to speculate that the extreme social structure in naked-mole rats might be related to the aversive state that is evoked by solitude – providing a strong enforcement signal for the animals to live in one giant colony where they are never alone. We have begun our behavioral mapping with somatosensory stimuli, but we will perform other tests to determine if other behaviors are altered during isolation. Perhaps isolation induces an entire state change, and altered somatosensory perception is just one of several systems affected. We plan to investigate this and other fascinating biology of the naked-mole rat as we open up this system to modern neuroscience approaches.

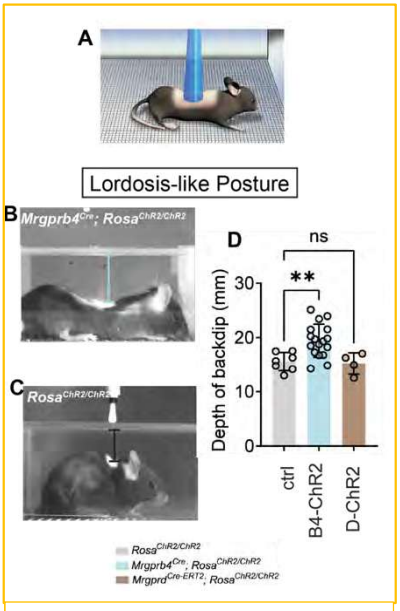


Figure 5: Focalized activation of Mrgprb4-lineage neurons in the back skin induces lordosis-like back lowering posture. A) Schematic illustrating light placement to shaven back skin and posture change. B) *Mrgprb4^{Cre}; Rosa$^{ChR2/ChR2}$* females exhibit a striking lordosis-like dorsiflexion in response to Mrgprb4-lineage neuron activation, a behavior absent from (C) *Rosa$^{ChR2/ChR2}$* female littermates. D) This posture is quantified as the back's maximum distance from the chamber ceiling over the course of 20s optogenetic stimulation. Note posture not observed when doing the same experiment with a related population of MrgprD neurons. Adapted from XXXXXXXXXXXXXXXXXX

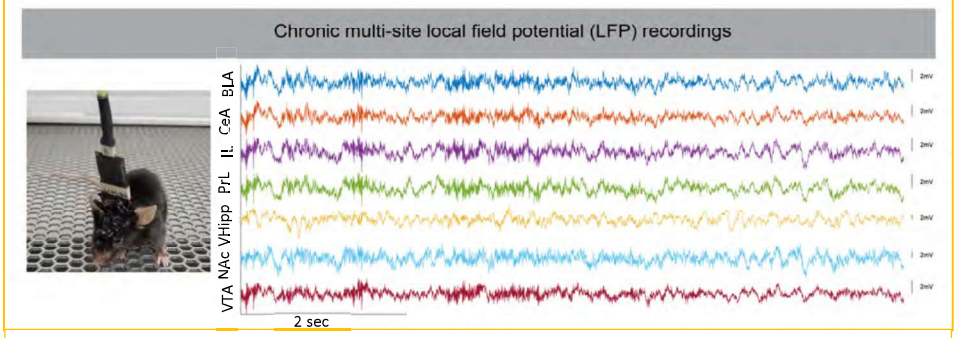


Figure 6. (Left) Chronic Local Field Potential recordings in freely behaving mice. (Right) Mice were implanted with electrodes targeting the infralimbic cortex (IL), prelimbic cortex (PrL), nucelus accumbens (Nac), ventral hippocampus (vHipp), ventral tegmental area (VTA), basolateral amygdala (BLA), and central amygdala (CeA). Adopted from XXXXXXXXXXXXXXXX.

Future Directions for the Next 5-10 Years

Future Direction 1: Identifying brain-wide activity signatures for chronic pain using awake fMRI and chronic multi-circuit electrophysiology coupled to high-speed videography of behavior

Here, we will test our hypothesis that chronic pain is not a monolith, but unique brain and behavioral signatures drive distinct states. In the same cohorts of mice (using both sexes of mice as well as young and aged mice), we will use our automated pain assessment platforms and on subsequent days we will perform daily resting state functional magnetic resonance imaging (fMRI) in mice before and after the animals have undergone one of three unique manipulations for inducing chronic pain including neuropathic injury, bone cancer, and knee-joint osteoarthritis. Although fMRI in mice for brain imaging is considered somewhat unorthodox and still underexplored as an approach, labs have been successful using this approach [20-22], and the upside is tremendous. Using fMRI in the awake mouse will allow us to use the brain-wide functional maps as a launchpad to gain mechanistic detail and test causal predictions at cellular resolution, which is impossible to do in humans. We are already underway beginning to perform these experiments in collaboration with our colleague XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX. After uncovering the functional connectivity maps with fMRI, we will employ causal based manipulations where we use viral injection strategies to perform optogenetics and chemogenetics to the brain to activate or inactivate functionally connected brain regions to determine if disrupting the signatures can block sensory or negative emotion-linked behaviors across our chronic pain models.

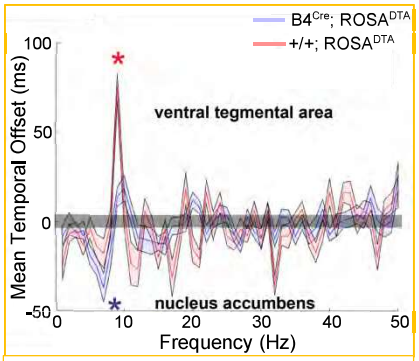


Figure 7. Ablation of Mrgprb4-lineage neurons alters LFP signal directionality in the VTA-Nac reward circuit. The phase offset time series was calculated between the VTA and Nac. The red asterisk indicates the frequency at which the VTA leads the NAc to a larger degree in control mice. Adopted from XXXXXXXXXXXXXX

We will also take a more biased and hypothesis driven approach in mice, with an equally technically challenging paradigm using in vivo multi-circuit neurophysiological recordings across seven brain regions simultaneously (**Figure 7**), coupled to real-time recording of behavior. We will test the hypothesis that the three chronic pain states tested above will alter network activity across brain regions previously implicated in negative emotion and depression (**Figure 7**). We will record local field potentials (LFPs), which are pooled electrical activity units of neurons, in these areas as we simultaneously measure behavioral responses to evoked or spontaneous bouts of pain. This experimental paradigm will allow us to "read" the animal's brain while mapping pain sensation and emotion at timescales ranging from milliseconds to minutes. With this data, we will calculate 1) phase offset to estimate signal directionality between each pair of brain regions, 2) LFP amplitude across frequencies, and 3) LFP synchrony between brain regions at millisecond timescales. Since related work has been performed in the mouse for major depression [23], we will also use the depression model in the prior study as a control group for comparison to our pain models. **In summary, we will be able to determine if chronic pain models share related network signatures to another negative emotion state or if pain unlocks an entirely novel and emergent set of network signatures.**

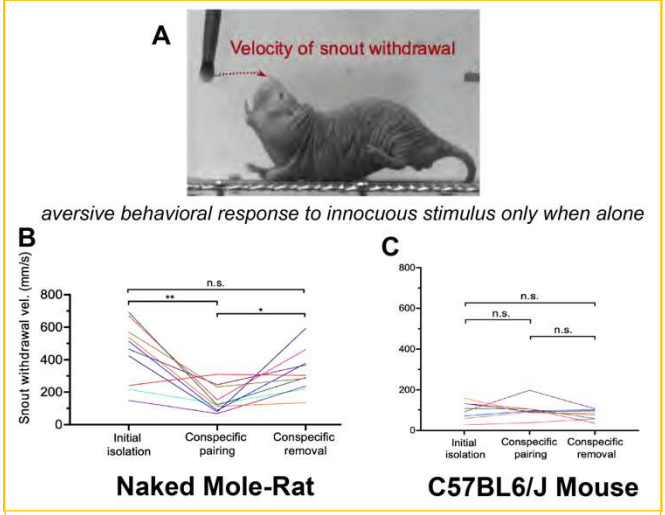


Figure 8. Naked mole-rats have aversive response to soft stimuli to the facial skin only in isolation. A) Upon stimulation to the snout, the stimulus barely touches the animal before it does a rapid withdrawal and screaming behavior. B) We measured snout withdrawal velocity as a proxy for pain or aversion and this is modulated by the placement or removal of a conspecific into the testing chamber. The results happen within minutes. N=10 naked mole-rats. C) The altering of somatosensation by presence or removal of a conspecific is not observed in standard lab mice. N=10 mice. Statistics are one-way ANOVA followed by Tukey's multiple comparisons test.

In addition to using mice in the fMRI scanner, we will also study naked mole-rats. In particular, we will ask, how does brain activity look in an animal challenged with inflammatory molecules in the periphery that it seems to ignore from a behavioral level. Naked mole-rats have similar numbers of nociceptors and these neurons fire to capsaicin in cell culture, although the animals have no behavioral response to this normally painful

molecule. In addition to peripheral alterations that have been discovered, such as polymorphisms in ion channels [8,24,25], and the lack of specific pain neuropeptides like substance P [26], ***what if the brain asserts a powerful descending inhibition system to block some forms of pain in naked mole-rats?*** Most mammals studied have strong descending pain inhibition systems [12], and if the naked mole-rat has evolved a new kind of system we could potentially discover it with our fMRI approach. Considering long-term impacts of such a discovery, if we identify a potent pain inhibition system in the naked mole-rat, this finding could potentially be harnessed as a new approach to pain relief.

Future Direction 2: Skin-brain interactions for rewarding social touch

Driven by our findings manipulating the Mrgprb4-lineage touch neurons, we want to build on this work and study other sensory neuron populations and other touch-dependent behaviors. Our prior studies focused on mating behavior (**Figure 4**), but other social behaviors – ranging from mothers stroking and licking pups to animals grooming one another, also activate touch neurons that feed into central circuits for social reward. The integrated network from skin-to-brain that drive these interactions remain obscure. **I hypothesize that molecularly-defined peripheral sensory neurons contribute to unique socially rewarding behaviors and we will design experiments to test this hypothesis.** My lab will study available sensory neuron cre driver lines [27,28], as well as make our own new sensory neuron cre-driver lines based on new sensory neuron single-cell sequencing papers [29], and test the functional consequence of activating or inhibiting various populations of touch neurons. At the periphery, we will develop a whole animal in vivo preparation to record calcium activity from defined sensory neuron touch populations in head fixed but awake animals as we deliver myriad somatosensory stimuli to the body. Moreover, we will develop an optogenetic-fMRI paradigm to determine which brain networks are driven by stimulating molecularly defined touch neurons in the periphery. These experiments will be coupled to viral strategies for connecting the periphery to the spinal cord and brain structures, as well as fiber photometry and micro-endoscope imaging, to determine cell-type resolved neural activity patterns driven by touch inputs (analogous to experiments performed in our recent study XXXXXXXXXX). Although we will begin these studies in the mouse because of the wealth of molecular and genetic tools, it is my expectation that we will also perform complementary studies in the naked mole-rat as we develop new tools and have new sequencing datasets to investigate unique cell types.

Future Direction 3: Developing naked mole-rats as a model to study somatosensation and social structures

To perform the types of causal based studies common in more standard model organisms – from gene manipulation to brain recordings, we will need to develop new paradigms listed below. While developing these paradigms we will continue to do behavioral mapping experiments and begin some causal based studies using traditional pharmacology approaches.

a. Develop paradigms to record from the awake naked mole-rat brain. We will attempt to use 1) neurotransmitter sensors for molecules such as oxytocin and dopamine with fiber photometry in freely behaving animals; 2) AAV based delivery of fluorescent calcium indicators recorded with micro-endoscopy in freely behaving animals; 3) 2-photon calcium imaging in head-fixed animals either alone or in a group while applying various sensory stimuli and imaging in requisite sensory cortices.
b. Comparative single-cell genomics using RNAseq and ATAC-seq of peripheral sensory ganglia, dorsal horn of the spinal cord, and pain/somatosensory brain regions, comparing naked mole-rats to other mammals where this gene expression data is already available. The gene expression signature of the naked mole-rat could provide fundamental clues about how evolution has shaped an animal to be insensitive to certain forms of pain. I anticipate finding gains or losses of genes/cell types known to drive pain in standard lab mice and rats, and such a finding could point us towards which target(s) to go after for blocking pain.
c. Develop viral strategies to target cell types and brain circuits in both the periphery and brain with indicators and actuators like opsins and DREADDs.

Summary

Since opening my lab in 2018, I have positioned myself as a productive investigator in the field of pain and somatosensation, and over the next 5-10 years I anticipate emerging as one of the field's leaders. In particular, I aim to be known for, **1)** providing new conceptual advances in our fundamental understanding of how pain is represented in the brain and how the sensory and emotional dimensions cooperate to generate perception – both in normal and chronic pain states. In addition, I aim to be known for, **2)** using the naked mole-rat to understand new fundamental biology about pain inhibition systems and how somatosensation and social structures converge to drive behavior.

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